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Capital One — Distinguished AI Engineer, Intelligent Foundations & Experiences (IFX)

Dear Capital One IFX Team,

The hardest problems in production AI aren't the models — they're evaluation, observability, and guardrails. Knowing whether the model is working when it matters. Catching drift before it reaches customers. Making AI decisions auditable enough to trust at financial services scale. That's the infrastructure IFX builds. That's also what I've spent the last two years building.

AgentProbe — github.com/venkatalakshmiparimala/AgentProbe — is a unified evaluation and observability platform for agentic AI: a **5-metric LLM-as-judge engine** (factual accuracy, hallucination, retrieval quality, completeness, compliance), CI/CD quality gates that block failing runs before production, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), and a React dashboard tracking per-run metrics, regression drift, and latency across concurrent agent test suites. I built it because I ran into every one of these problems in production and decided to solve them systematically.

The production grounding came from Entrata. I shipped the company's first AI pipeline — Python, PyTorch, OpenAI API — across **1.6M real financial records**, with output validation, quality gates, and compliance controls in place before a single record went live. I also achieved **100% PHI compliance** on an AWS S3 migration with Libsodium-encrypted pipelines (zero security incidents) and built data pipelines across S3, EC2, RDS, and Glue for 1M+ property records at 99.5% uptime. Financial services data environments — the rigor, the compliance requirements, the zero-tolerance for errors — are not unfamiliar to me.

Leasel adds the eval and auditability layer: an agentic LangGraph RAG pipeline with **93.6% precision and 87.9% F1** across 10 compliance rules, a pass@k eval harness (pass@1 81.7%, pass@3 88.3%), and a deterministic rules engine that makes every decision fully traceable. Building AI for regulated domains taught me that "mostly right" is a liability, not a feature.

IFX is building the platform layer that the rest of Capital One's AI runs on. Getting that infrastructure right — evaluation, observability, guardrails, LLM routing — determines the quality of every product built on top of it. I want to help build it, and I'm local to McLean.

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